

Part 1 — Dataset selection

Gene: FtsZ

Organisms:

```
Escherichia coli
Pseudomonas aeruginosa
Bacillus subtilis
Caulobacter crescentus
```

Novelty: I have used a different set of genes from bacteria. I used different bacteria than I used in the previous class examples.

Part 2 — BLAST analysis

BLAST program: blastn

Two species selected:

```
Escherichia coli
Pseudomonas aeruginosa
```

Interpretation:

The e-value is very small ($7e-75$). This means that the match is significant. The chance of this alignment happening by random is basically zero. The query coverage is 79%. This means that most of *Escherichia coli*'s sequence aligns with *Pseudomonas aeruginosa*. The percent identity is 67.69%. This means that roughly 68 percent of the nucleotides are exactly the same. This means that the sequences are very similar but not identical.

Limits:

One conclusion I can make that the match is statistically significant. The low e-value and the other matches show that the FtsZ gene in both bacteria are similar. They possibly have the same coding region that allows it for cell division. However, it might work a little different depending on the organism. One conclusion I can't make is homology. One could possibly not determine homology with BLAST. It cannot determine evolutionary relationships between species.

BLAST screenshot:

Activity 05 - BLAST

Section Report 1 — Sequence Similarity, Alignment, and Phy...

BLAST

NCBI Blast:Nucleotide Sequence

Job Title

Nucleotide Sequence

RID

TVVCEAHG014

Search expires on 02-26 01:27 am

Download All

Program

BLASTN

Citation

Database

core_nt

See details

Query ID

lcl|Query_6170859

Description

None

Molecule type

dna

Query Length

1152

Other reports

Distance tree of results

MSA viewer

Filter Results

Organism

only top 20 will appear

exclude

Type common name, binomial, taxid or group name

Add organism

Percent Identity

E value

Query Coverage

to

to

to

Filter

Reset

Descriptions

Graphic Summary

Alignments

Taxonomy

Sequences producing significant alignments

Download

Select columns

Show

100

select all

100 sequences selected

GenBank

Graphics

Distance tree of results

MSA Viewer

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	Pseudomonas aeruginosa strain 2020CK-00220 chromosome, complete genome	Pseudomonas aeruginosa	289	289	79%	7e-75	67.69%	6483169	CP124648.1
☒	Pseudomonas aeruginosa strain LS.2c chromosome, complete genome	Pseudomonas aeruginosa	289	289	79%	7e-75	67.69%	6349540	CP092634.1
☒	Pseudomonas aeruginosa strain 85505 chromosome, complete genome	Pseudomonas aeruginosa	289	289	79%	7e-75	67.69%	6426922	CP173142.1
☒	Pseudomonas aeruginosa strain PaLo527 chromosome, complete genome	Pseudomonas aeruginosa	289	289	79%	7e-75	67.69%	6283865	CP075778.1
☒	Pseudomonas aeruginosa strain 2021CK-01851 chromosome, complete genome	Pseudomonas aeruginosa	285	285	79%	3e-73	67.58%	6440143	CP124654.1
☒	Pseudomonas aeruginosa strain AR_0353 chromosome, complete genome	Pseudomonas aeruginosa	285	285	79%	3e-73	67.58%	7240677	CP027172.1
☒	Pseudomonas aeruginosa strain PaLo530 chromosome, complete genome	Pseudomonas aeruginosa	285	285	79%	3e-73	67.58%	6761200	CP075773.1
☒	Pseudomonas aeruginosa strain 2021CK-01494 chromosome, complete genome	Pseudomonas aeruginosa	285	285	79%	3e-73	67.58%	6583426	CP124669.1

Part 3 — Multiple sequence alignment

Aligner: tcoffee

Justification:

I chose tcoffee over clustal because in my opinion, it provides a more accurate alignment. I believe it handles and places gaps more accurate. It doesn't force alignments like clustal does.

DNA observations:

The DNA alignment between bacteria share some long stretches of similarity, however there are a few differences in some regions. Around 150–200, you can see multiple single-base substitutions between E. coli and P. aeruginosa. Larger gaps appear around 300–350. You can also see that Caulobacter crescentus has extra inserted bases that the other species don't have. Another area that's different is around 900–950, where there are scattered mismatches and small deletions.

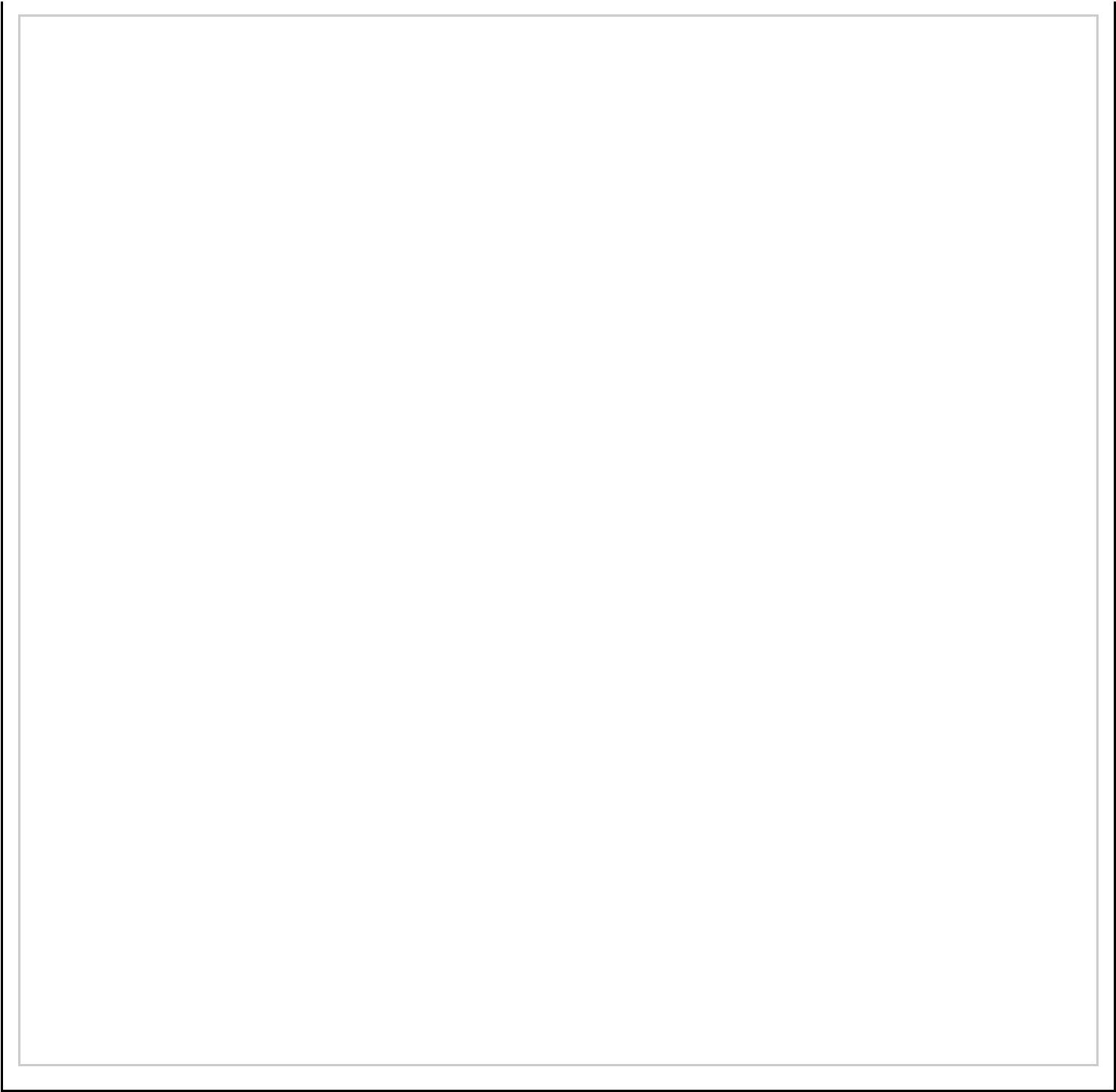
Protein observations:

The protein alignment, most of the conserved region's cluster are around 1–200. Differences and gaps start to appear and become more obvious near end around 300–370. *C. crescentus* has extra amino acids around 330–360, which the other bacteria don't have. Small insertions and more mismatched amino acids also show up around 250–280.

NESPrpt DNA PDF:



ESPrpt Protein PDF:



Part 4 — Phylogenetic analysis

Tree method: -

Method explanation:

Maximum Likelihood and Neighbor-joining are both methods of making a phylogenetic tree. The difference between the two is that Maximum likelihood uses a statistical/character approach while Neighbor-joining uses a distance approach. They make different assumptions.

Bootstrap explanation:

Bootstrapping is used to test the reliability of a phylogenetic tree. It helps support the relationship between different species.

Bootstrap interpretation:

A higher bootstrap means that the relationship is consistently recovered while a low bootstrap value means the relationship is uncertain. My bootstrap value was 99.5% means that the relationship between the species is very strong and certain.

Part 5 — Rooting

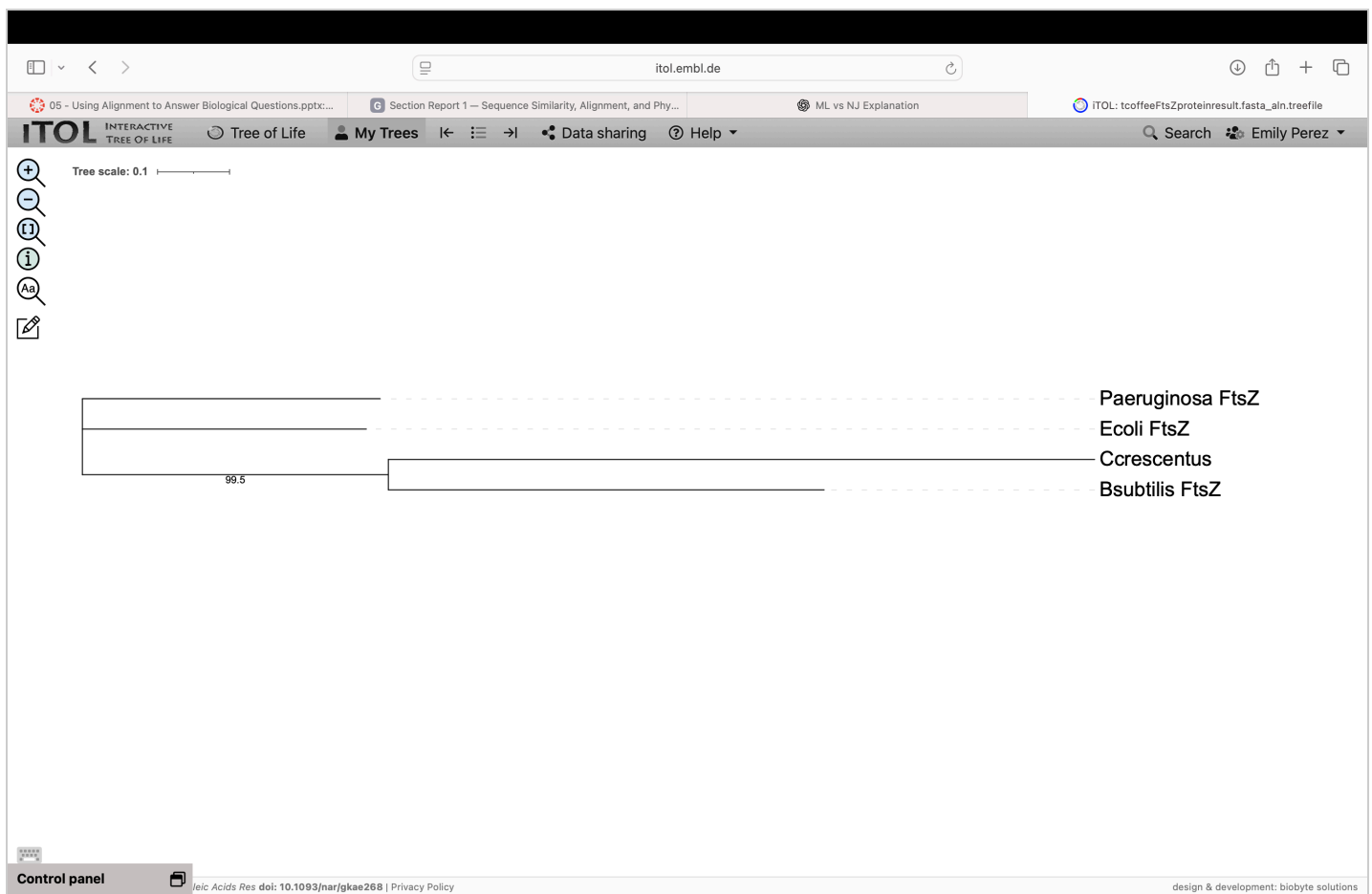
Unrooted vs midpoint-rooted comparison:

After midpoint rooting the tree, I noticed that some species were moved around to show that they are more closely related than the other ones.

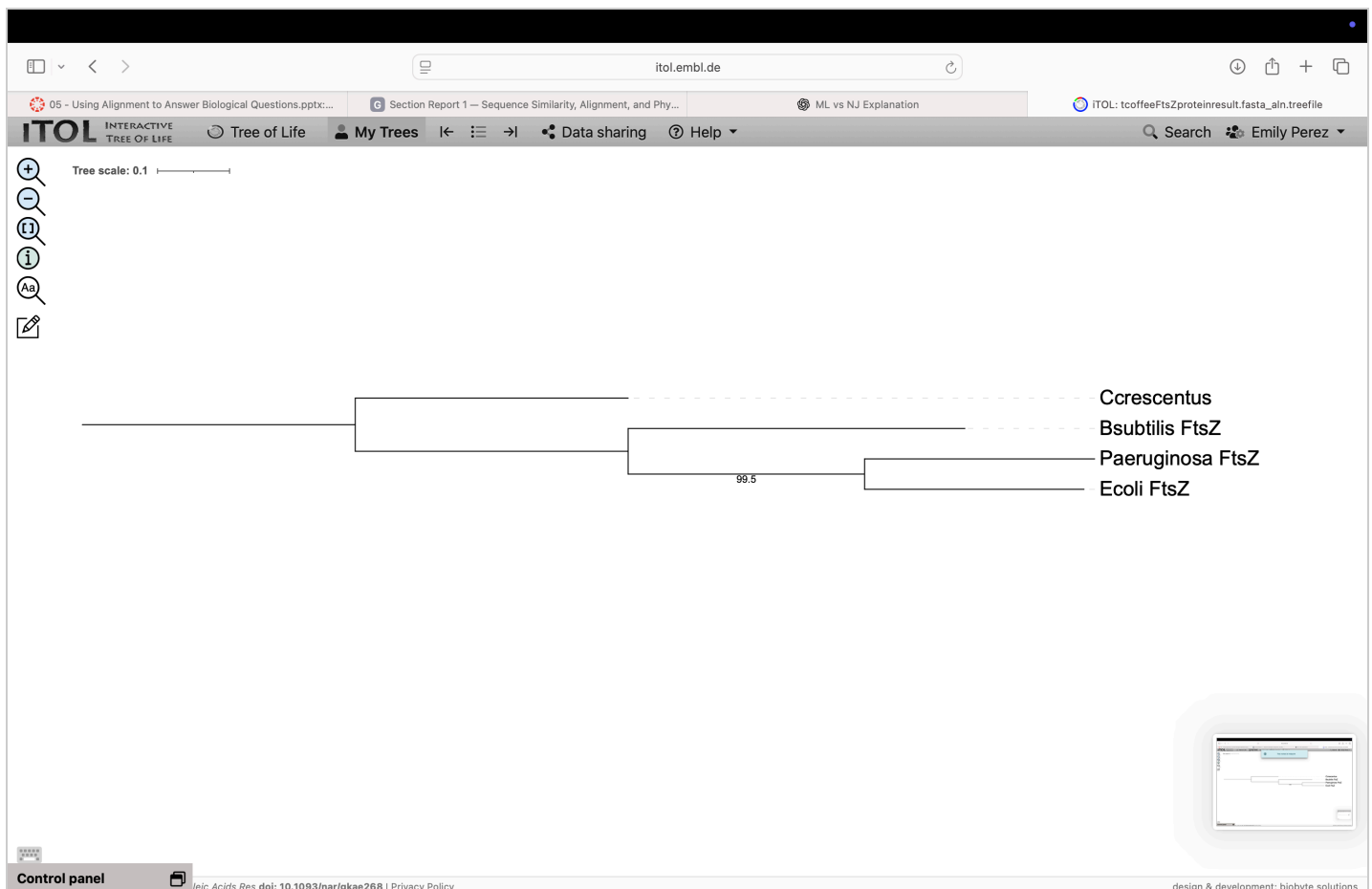
Midpoint rooting assumption:

Midpoint rooting allows us to think that the species have the same common ancestor. That they all evolved at the same rate. Using an outgroup needs other biological information, midpointing is strictly using the data only.

iTOL Unrooted screenshot:



iTOL Midpoint-rooted screenshot:



Part 6 – Synthesis

Overall, doing this type of work helps me understand the evolutionary relationship between different species. I can do this by looking at the different types of DNA and protein alignments or plugging them into the phylogenetic tree and seeing it that way. I believe uncertainty can be introduced early on when running the multiple alignments. Using tcoffee or cluster. One software is a bit better than the other because it doesn't seem to force alignments. Choosing the better software and knowing the difference is important for the uncertainty to be mitigated.